

Project Information

The final project is a chance to show what you have learned in class and to create an example of both your coding and communications skills that you can show to employers and admission committees. The final deliverable is made up of two parts:

1. **An executive summary of your analysis.** This is a professional report to be turned in as a PDF. This report clearly lays out the question(s) you are addressing, discusses the analysis you performed in the notebook, and draws a conclusion. This report should be no more than three pages long (1,500 words, not including figures and tables). This is not a technical document, but you should discuss any technical difficulties and any assumption you made in your analysis.
2. **Code and data.** You will submit a Jupyter notebook and any other files needed to make the code work. This notebook should explain what you are doing (and have many useful comments) but does not contain your questions/conclusions. Think of this as the document you would give a coworker who would like to replicate your results and understand what you have done to get to those results.

Note: You will need to state where you got your data. "Canned" datasets from sources like Kaggle and other analytics sandboxes are not allowed without prior approval.

Teams

The project can be completed in teams of two to four students. Students working in teams should not divide up the work so that one person does all the coding, one person does all the writing, etc. Each team member should contribute to all parts of the project.

Each team should choose a corresponding author. This person will submit documents through Canvas. The other team members do not need to submit anything.

Deliverables

A survey response with your project *roster*, the members of your team, and a proposed topic area

Proposal

The proposal is one-page document (PDF) and the accompanying Jupyter notebook and data files. Upload them as **two files**:

1. The proposal document as a single PDF

2. The code and data in a single zip file

The proposal **document** should include

1. the questions your team plans to answer
2. the data you will use to answer them
3. one graph or table that demonstrates a pattern in your team's data
4. (be sure to list your team members and note both your team number and also which member is the corresponding author)

The **Jupyter notebook and data files** should load the data and produce the graph or table that is in the proposal document.

Note that your team does not need to have answers for your questions yet. The goal of the proposal is to come up with a set of questions and the appropriate data. The table or graph is to let us know that you all have figured out how to get the data into pandas and that your team has looked through the data.

Your questions and expected results will likely change a bit as you study the data more and learn new things. That is how research works. The proposal is to ensure that we agree that the project is off on the right foot.

Final project files

You must upload two files to canvas. **Upload them as two files.** Do not compress your project report.

1. The project report as a PDF. Do not forget to include your team number and the names of your group members in the report.
2. The code and data in a zip file
 - The code and data should be in one folder that you zip to create a single zip file.
 - Before you zip your code, make sure that there are **no absolute file paths**. If your code is in a folder on your computer named 'project' and the data file you want to read in is in a folder named 'data', your code should **not** have paths such as `C:\Users\jbrauer\Documents\project\data\file.csv`. All the paths should be relative to the folder your notebook is in, such as `data\file.csv`.

Templates

The proposal templates are MS Word documents. Students interested in academic research might want to try preparing the documents in **LaTeX** [↗\(https://www.latex-project.org/\)](https://www.latex-project.org/), which is the standard in technical writing. **Overleaf** [↗\(https://www.overleaf.com/\)](https://www.overleaf.com/) is an easy-to-use online LaTeX editor.

Word templates

[Proposal template \(https://canvas.wisc.edu/courses/502185/files/50161627?wrap=1\)](https://canvas.wisc.edu/courses/502185/files/50161627?wrap=1) 
(https://canvas.wisc.edu/courses/502185/files/50161627/download?download_frd=1) // [Project template \(https://canvas.wisc.edu/courses/502185/files/50161620?wrap=1\)](https://canvas.wisc.edu/courses/502185/files/50161620?wrap=1) 
(https://canvas.wisc.edu/courses/502185/files/50161620/download?download_frd=1)

Latex templates

[Project template \(https://canvas.wisc.edu/courses/502185/files/50161643?wrap=1\)](https://canvas.wisc.edu/courses/502185/files/50161643?wrap=1) 
(https://canvas.wisc.edu/courses/502185/files/50161643/download?download_frd=1)

Grades

Projects will be graded on both the quality and readability of the code and the project report. The grading rubric is [here \(https://canvas.wisc.edu/courses/502185/pages/final-project-grading-rubric\)](https://canvas.wisc.edu/courses/502185/pages/final-project-grading-rubric). Some of the things we will be looking for include:

1. Project idea: Is the question clearly stated? Are the data appropriate? Is the approach appropriate?
 2. Code: Does it run? Is it commented? Is it readable? Are complicated parts of the code explained?
 3. Project report: Is the report clearly written? Are the graphs labeled? Do they display graphical excellence?
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Generating ideas

Many coding or statistics classes that require projects assign project ideas to students. Not here! Part of the data mentality that we are developing is being able to ask good questions. Like any skill, this takes some practice.

- Read. Blogs, newspapers, magazines...almost anything can spark a question. Reading publications that are not afraid to show some data are even better: The Wall Street Journal; New York Times; Economist; FiveThirtyEight...
- Think about other classes you have taken. What questions came up in labor class? Industrial organization? Sociology?
- What interests you outside of economics? Sports, theater, food? What questions and data are out there?

- Keep an idea journal. Ours are text files on our computers. Anytime an idea strikes, we write it down. Periodically, we go over the journal to see what questions we might be able to make some progress on.
 - The [course resources page \(https://canvas.wisc.edu/courses/502185/pages/data-resources\)](https://canvas.wisc.edu/courses/502185/pages/data-resources) is the beginning of a list of data sources and writing about data. Send us more ideas!
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Examples

There are numerous resources online that are similar in format of your projects

- Policy briefs on health care topics at [1% Steps for Health Care Reform](https://onepercentsteps.com/)  (<https://onepercentsteps.com/>)
- Short, non-technical summaries of full-length economics papers: [NBER Digest](https://www.nber.org/digest)  (<https://www.nber.org/digest>); [Microeconomic Insights](https://microeconomicinsights.org/)  (<https://microeconomicinsights.org/>)